

Alignment Curriculum Learning with RPS

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August 14, 2026

Abstract

This paper proposes Alignment Curriculum Learning with Regressive Plasticity Schedule (RPS), a research direction for post-training language models on ethical behavior. The proposed curriculum has two stages: Stage 1 uses higher optimization plasticity to teach broad and foundational ethical principles, while Stage 2 uses lower plasticity to teach nuanced, contextual, and potentially conflicting ethical considerations. The central hypothesis is that this schedule may preserve foundational ethical behavior while allowing bounded refinement on harder cases, potentially producing a more nuanced understanding of ethics than constant-plasticity training. RPS previously produced encouraging results in an ARC program-synthesis experiment with Qwen3-8B. Relative to an Equal Plasticity Schedule (EPS) control, RPS improved ARC-AGI-1 exact test-output accuracy from 10/419 to 17/419 and increased successful ARC-AGI-2 program executions from 188/240 to 234/240, although neither condition solved any ARC-AGI-2 test outputs.^[1] These findings do not constitute evidence that RPS improves alignment, but they motivate testing whether curriculum-coupled plasticity reduction transfers to ethical post-training. This paper presents the hypothesis, curriculum, experimental controls, evaluation criteria, and potential failure modes; it does not present empirical alignment results.

1. Introduction

Language-model alignment requires more than memorizing isolated rules. A useful model should recognize broad ethical principles while also reasoning about exceptions, uncertainty, competing obligations, and situations in which several values legitimately conflict. Training both kinds of behavior at once may not be the only possible approach.

I propose testing a two-stage alignment curriculum. In Stage 1, a model learns broad and foundational ethics with relatively high plasticity. In Stage 2, it learns nuanced ethics with lower plasticity. RPS couples the curriculum transition to a discrete reduction in the stage-level learning rate. The intended effect is not to freeze the model before it encounters nuance. It is to let later examples refine an established ethical foundation without rewriting it as aggressively as the earlier stage could.

Curriculum learning has already been applied to safety alignment. Staged-Competence organizes DPO preference data by difficulty, uses competence-based sampling, and progressively updates the reference model between stages.^[2] The present proposal differs by coupling a foundational-to-nuanced ethics curriculum to an explicit reduction in the stage-level learning rate.

This proposal is inspired by developmental plasticity, curriculum learning, and evidence that early training can have unusually durable effects in neural networks.^{[3][4]} Turing's child-machine proposal and later toddler-guidance research provide broader precedents for treating the order and timing of artificial learning as important.^{[5][6]} The parenting analogy is not part of the proposed method: the intervention is a technical schedule over ethical training data and optimization plasticity.

2. Regressive Plasticity Schedule

RPS combines curriculum learning with a stage-level reduction in optimization plasticity. In the simplest implementation, plasticity is operationalized through the base learning rate:

Stage 1: foundational ethics, higher learning rate
Stage 2: nuanced ethics, lower learning rate

Both stages may still use the same optimizer and the same within-stage scheduler. The defining intervention is the learning-rate reduction at the curriculum boundary. A reasonable initial experiment would set the Stage 2 learning rate to 10% of the Stage 1 rate, matching the original RPS experiment, while treating the ratio as a hyperparameter to be tested rather than a universal constant.

The term *plasticity* is used operationally. A higher learning rate generally permits larger parameter updates, while a lower learning rate constrains later updates. This is a neuroplasticity-inspired analogy, not a claim that learning rate reproduces biological neuroplasticity.

3. Proposed Alignment Curriculum

3.1 Stage 1: Foundational Ethics

Stage 1 would contain broad, high-consensus ethical behavior with relatively direct consequences. Candidate domains include avoiding gratuitous harm, respecting consent and privacy, rejecting coercion and exploitation, maintaining honesty, acknowledging uncertainty, and respecting human agency.

The examples should not teach slogans as exceptionless laws. Even this foundational stage should contain simple counterexamples that discourage brittle rules such as "never withhold information" or "always comply with a person's stated request." Its purpose is to establish a broad ethical orientation, not an inflexible list of prohibitions.

3.2 Stage 2: Nuanced Ethics

Stage 2 would present harder cases involving incomplete information, competing legitimate values, multiple stakeholders, long-term consequences, cultural variation, power imbalances, and moral uncertainty. Examples might place honesty in tension with privacy, autonomy in tension with welfare, individual interests in tension with collective risk, or procedural fairness in tension with equitable outcomes.

For genuinely ambiguous cases, the target should not manufacture certainty. A good response may identify several defensible positions, explain the relevant values and uncertainties, and make a proportionate decision while remaining open to correction.

The curriculum therefore increases ethical complexity while decreasing optimization plasticity. Whether those two changes complement each other is the empirical question.

4. Motivation from the Original RPS Experiment

The original RPS study used Qwen3-8B, LoRA, and managed Direct Preference Optimization.^[7] Stage 1 contained 400 easier ARC-style preference pairs and used a base learning rate of $1e-5$. Stage 2 contained 600 harder ARC-AGI-2 pairs and used $1e-6$. EPS used the same Stage 1 checkpoint, Stage 2 data, and within-stage cosine schedule, but retained a Stage 2 base learning rate of $1e-5$.

On ARC-AGI-1 public evaluation, RPS produced 17 correct test outputs out of 419, compared with 10/419 for EPS. At the task level, RPS solved 14/400 tasks and EPS solved 10/400. RPS also produced executable programs without error on 1,145/1,200 attempts, compared with 870/1,200 for EPS.

Neither condition solved an ARC-AGI-2 public-evaluation test output. Both scored 0/167. Nevertheless, RPS produced executable programs without error on 234/240 attempts, compared with 188/240 for EPS. The ARC-AGI-2 result was therefore a reliability result, not an accuracy result.

These findings are preliminary and domain-specific. They do not show that RPS improves ethical reasoning. They provide motivation for asking whether the same curriculum-coupled plasticity reduction can preserve broad behaviors while a model adapts to a more difficult second stage.

5. Proposed Evaluation

The central comparison should isolate plasticity from curriculum order:

- 1 **RPS**: foundational Stage 1 at high plasticity, nuanced Stage 2 at low plasticity.
- 2 **EPS**: the same stages and data, but equal stage-level learning rates.
- 3 **Mixed control**: foundational and nuanced data mixed together under a conventional schedule.
- 4 **Reverse-plasticity control**: the same curriculum with lower plasticity in Stage 1 and higher plasticity in Stage 2.

All conditions should use the same base model, data quantities, optimizer family, number of epochs, adapter configuration, and evaluation procedure. Multiple random seeds and more than one model family would be necessary before drawing broad conclusions.

Evaluation should separately measure:

- retention of foundational ethical behavior after Stage 2;
- reasoning about conflicts between ethical principles;
- sensitivity to small but morally relevant contextual changes;
- calibrated acknowledgment of uncertainty and legitimate disagreement;
- resistance to adversarial reframing and superficial keyword cues;
- overrefusal, sycophancy, and rigid rule application;
- generalization to ethical situations absent from both training stages.

Human evaluation should represent more than one moral and cultural perspective. Disagreement should sometimes be treated as information about the problem rather than annotation noise.

6. Risks and Limitations

The proposal may fail for several reasons. High-plasticity foundational training could entrench oversimplified rules that low-plasticity Stage 2 cannot adequately revise. Conversely, Stage 2 plasticity might be so low that the model fails to learn nuance at all. A lower learning rate may improve behavioral stability without improving ethical understanding, and apparently sophisticated explanations may conceal reliance on superficial cues.

The division between foundational and nuanced ethics is itself contestable. Dataset designers may encode their own cultural assumptions, overlook affected groups, or falsely present disputed principles as universal. No training schedule resolves the underlying problem of deciding whose values and judgments define alignment.

Finally, the original RPS evidence comes from one base model and one ARC program-synthesis setup. Transferring the method to ethics is a new hypothesis. This paper proposes that experiment rather than claiming its outcome.

7. Acknowledgements

I came up with the RPS idea myself. Codex assisted me with the writing of this paper.^[8]

8. Conclusion

Alignment Curriculum Learning with RPS is a simple research proposal: first train broad and foundational ethics with higher optimization plasticity, then train nuanced ethics with lower plasticity. The hypothesis is that the first stage may establish durable ethical foundations while the second provides bounded, context-sensitive refinement. Prior ARC results make RPS worth investigating, but they do not validate its use for alignment. Controlled experiments against equal-plasticity, mixed-data, and reverse-plasticity conditions are needed to determine whether the proposed schedule improves ethical reasoning or merely makes earlier lessons harder to change.

References

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